



Thermal Noise in Track-and-Hold Circuits

Analysis and simulation techniques.

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Switched-capacitor (SC) circuits are ubiquitous in today's complementary metal-oxide-semiconductor (CMOS) mixed-signal ICs and are used primarily in filters, data converters (A/D and D/A), sensor interfaces, and dc-dc converters. While the general idea of SC-based resistor emulation dates back to Maxwell [1], the adoption of analog sampled data circuits had to wait until MOS technology appeared as a viable process technology platform in the 1970s [2], [3]. Since then, an abundant amount of literature has been created on how SC circuits should be engineered in modern CMOS. Surprisingly, however, little attention has been paid to explaining the effects of thermal noise in SC circuits in a practically accessible manner. Therefore, in the same spirit as the recent paper by Schreier et al. [4], this tutorial article attempts to fill in some of the blanks that may persist in the minds of the SC circuit practitioner. The material presented herein draws from a tutorial held

at the 2011 International Solid-State Circuits Conference (ISSCC) [5].

As a vehicle that lets us explore the main ideas of this article, we will utilize the commonly used charge-redistribution track-and-hold amplifier (THA) circuit (shown in Figure 1) and evaluate this circuit in terms of its thermal noise performance. Of course, it is well known that the thermal noise of this circuit is somehow related to " kT/C ." What we accomplish in this article is to precisely track down the proper proportionality constants so that we can validate the results of a circuit simulation with good precision. After all, as you have learned from your experienced analog guru, you will never trust a circuit simulation blindly, right?

Thermal Noise and the Elementary Track-and-Hold Circuit

The word "noise" tends to be overloaded in engineering and describes a plethora of phenomena that cause some form of signal degradation. Many of these issues, such as

supply and substrate coupling, can be mitigated using proper engineering and will usually not prevail as the main limitation. This is typically also true for flicker noise, which can be minimized using a variety of clever design techniques [6]. At the end of the day, the main constraint on how accurately we can process signal samples in SC circuits is set by thermal noise, which is therefore the focus of our discussion.

Introductory material on the origins of thermal noise is found in most analog IC textbooks [7] and a variety of journal articles [8]. All we need to know for our purpose is that 1) thermal noise is associated with any resistive element, 2) it is uniformly distributed in frequency, and 3) it comes with a spectral density of $4kTR$, where k is the Boltzmann constant, T stands for absolute temperature, and R is the resistance value.

To investigate the impact of thermal noise in SC circuits, it is best to begin with the simplest possible configuration, which is the elementary track-and-hold circuit (see Figure 2). It consists of a single switch and capacitor, which tracks the input voltage when the switch is on and holds the end value during the off period. As shown, we are representing the MOS switch using an idealized model that includes only the device's on-resistance (R); the off-resistance is assumed to be infinite. This approximation has no significant bearing on the derived results and lets us focus more clearly on the primary effects. A more important parameter is the number of time constants (N) that fit into the tracking period. Precision SC circuits are typically designed with $N = 7 \dots 10$ to ensure accurate transient settling.

Now, to analyze the noise performance, we set the input to zero and add the noise source associated with the switch's on-resistance (see Figure 3). As visualized, the noise voltage will disturb the output through the circuit's RC network whenever the switch is on. However, when the switch turns off, the state of the

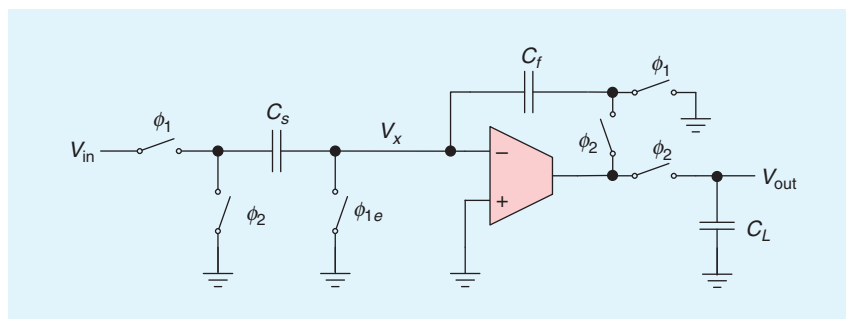


FIGURE 1: Charge-redistribution track-and-hold circuit. The circuit is drawn single ended for simplicity; actual implementations are usually fully differential.

capacitor is “frozen” and remains constant until the next turn-on. How can we describe this output waveform in a meaningful way?

To answer this question, we must first make a clear decision about how we are going to “look” at the output. One option is to treat the signal as a continuous-time waveform. However, this approach is not compatible with the common framework for SC circuit analysis, which relies on discrete-time signal abstractions. Therefore, what we are really looking for is the noise in the circuit's discrete-time output samples, indicated by the blue dots in Figure 3. Only two properties suffice to describe this discrete-time noise process: its total power (or, in other words, the variance of its sample distribution) and its power spectral density (PSD), which tells us how the power is distributed in frequency.

To find the total noise power, textbooks commonly invoke a variant of Parseval's theorem. From this theorem, it follows that the total power of the samples must be equal to the PSD of the thermal equilibrium noise process integrated over all frequencies. Mathematically, this means that we can obtain the total power by integrating the RC filtered version of the switch's thermal noise over all frequencies. This gives

$$\begin{aligned} \overline{v_{out}^2} &= \int_0^\infty 4kTR \left| \frac{1}{1 + j2\pi fRC} \right|^2 df \\ &= 4kTR \frac{1}{4RC} = \frac{kT}{C}. \end{aligned}$$

Interestingly, this famous result can be also be derived using much less intimidating math and by invoking the so-called equipartition theorem from statistical mechanics [8]. The equipartition theorem says that any energy storage element (or “degree

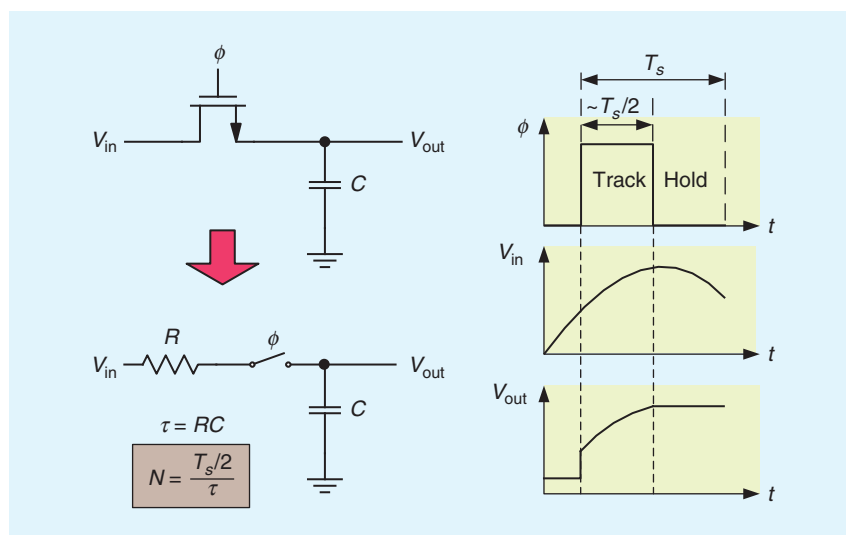


FIGURE 2: Elementary track-and-hold circuit with its first-order model.

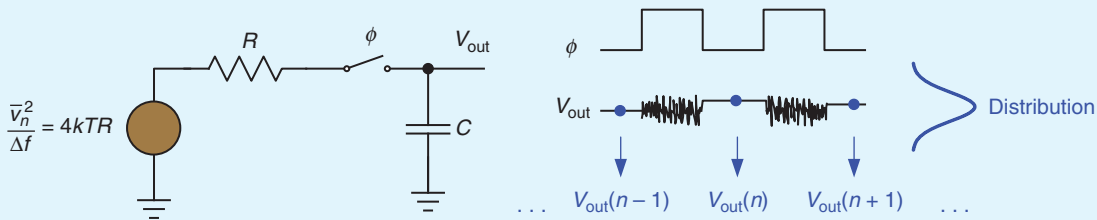


FIGURE 3: Elementary track-and-hold circuit with thermal noise source.

of freedom”) in thermal equilibrium holds an average noise energy of $kT/2$. In the circuit considered here, the capacitor is the energy storage element in question, and we can therefore write

$$\frac{1}{2} C \overline{v_{out}^2} = \frac{kT}{2}$$

$$\overline{v_{out}^2} = \frac{kT}{C}.$$

As we shall see later, the equipartition theorem is not only useful to confirm this basic result but will also help us in the analysis of the more complex THA circuit.

For those who are interested in the fine print, it is worth repeating that both of the above derivations assume that the circuit operates in “thermal equilibrium.” This leaves room for speculation on whether the kT/C bound can be overcome using techniques that would trick a circuit into nonequilibrium. Since this is indeed possible, we’ll spend a couple of minutes thinking about this from a high-level perspective, using a glass of iced tea as an analogy. First, note that for iced tea, reaching thermal equilibrium means that we let it warm up until it reaches the temperature of its surroundings. So, at least for this case, we know that one way to stay away from equilibrium is to add extra ice every now and then. Adding (dry) ice to our circuits would, of course, also help, but there are more sophisticated ideas. As shown, e.g., in [9], it is possible to align (via feedback) the state of a signal storage capacitor with that

of a larger capacitive “reservoir,” thereby achieving the equivalent effect to cooling.

Another idea would be to leave the tea in the fridge for most of the time, and take it out only for short periods of time that won’t let us reach equilibrium. This is almost exactly what can happen in SC circuits that are based on incomplete transient settling or switched current ramps; see, e.g., [10], [11]. Here, a certain amount of thermal noise is rejected simply because a subset of the noise sources are “on” for a duration shorter than the time constants in their path to the capacitive storage elements. While these and other ideas are very intriguing, they have not (yet) revolutionized electronics and often yield only incremental improvements over the equilibrium result when all thermal noise sources are considered.

Now that we have a handle on the total power of the discrete-time noise process, we need to think about its PSD. From basic random process theory, we know that one possible extreme is a “white” process, which implies that in the time domain there is no correlation from noise sample to noise sample. Another extreme would be a process in which the samples are highly correlated, leading to a shaped PSD that has the majority of the power concentrated at low frequencies, for example. In the circuit considered here, the determining factor that positions us within this set of possibilities is the length of the tracking phase relative to the circuit’s RC time constant, defined as N above.

For large N , the noise has a better chance to “destroy” and randomize the noise that was stored on the previous sample. Therefore, we expect the noise to turn white as N approaches infinity. Indeed, as shown in detail in [5], the PSD is given by

$$\text{PSD}(f) = \frac{2}{f_s} \frac{kT}{C} \frac{1 - e^{-2N}}{1 - e^{-2N} \cos\left(2\pi \frac{f}{f_s}\right) + e^{-2N}}$$

$$\approx \frac{2}{f_s} \frac{kT}{C} \text{ for } N > 3.$$

Therefore, for significantly large N factors used in practical circuits, the noise process’s PSD is essentially white and spreads its total power of kT/C uniformly across the range of discrete-time frequencies between 0 and $f_s/2$. From this result, it is then interesting to compare this result with the uniform PSD of the switch resistance

$$\left(\frac{2}{f_s} \frac{kT}{C}\right) / 4kTR = \frac{T_s/2}{RC} = N.$$

Thus, we see that the sampled noise PSD is N times larger than the PSD of the switch resistance by itself. This penalty can be explained by the aliasing effect that folds high-frequency components of the RC filtered noise back into the discrete-time band between $0 \dots f_s/2$ [12].

Ideal THA Operation

Now that we have a good handle on the basics of sampled noise analysis, we can turn our attention back

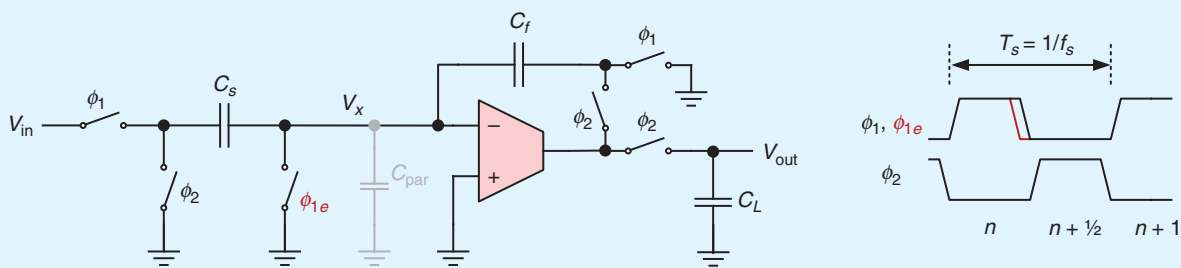


FIGURE 4: Detailed schematic of the charge-redistribution track-and-hold circuit.

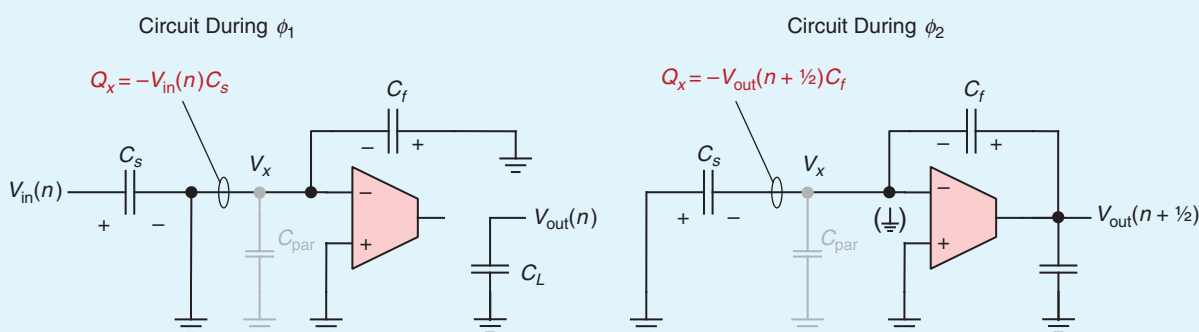


FIGURE 5: Charge-redistribution track-and-hold circuit in its two clock phases.

to the more complex THA circuit that we set out to analyze (Figure 1). However, before throwing noise into the mix, we should first understand how the circuit works under ideal and noiseless conditions, with perfect switches and using an ideal amplifier. For further investigation, we consider the circuit of Figure 4, which includes a few more details such as parasitic capacitance (C_{par}) at the amplifier input and a specification of the clock timing.

As shown, we assume that the THA is controlled by two nonoverlapping clock phases, ϕ_1 and ϕ_2 . To keep track of signal interaction between the clock phases, we assign a discrete-time index to each variable of interest. For instance, $V_{\text{in}}(n)$ is the input at the end of clock phase n , while $V_{\text{in}}(n + 1/2)$ is the signal one-half clock cycle later. Furthermore, we assume for simplicity and throughout this article that all settling transients have died out at these discrete-time instants; this is a reasonable approximation for the purpose of our investigation. How-

ever, one important timing aspect that we'll take into account later is that the switch labeled ϕ_{1e} turns off slightly before the falling edge of ϕ_1 . This technique is known as bottom plate sampling and helps minimize signal-dependent charge injection [13], [14]. Essentially all modern SC circuits with moderate-to-high precision rely on this trick to attain good signal fidelity.

For closer inspection, we now redraw the circuit in its two phases as shown in Figure 5. During the tracking phase (ϕ_1), the input voltage is acquired on the sampling capacitor C_s , while the feedback capacitor C_f is reset. In this phase, the sum of all charges on the capacitor plates connected to V_x is equal to $-V_{\text{in}}C_s$ and therefore determined by the input voltage. In the redistribution phase, (ϕ_2), the total charge at V_x changes to $-V_{\text{out}}C_f$, since now C_s and C_{par} (the parasitic capacitance at node V_x) are shorted via the virtual ground of the feedback amplifier. Now, since the total charge must be

preserved in the transition between the two phases, we can write

$$-V_{\text{in}}(n)C_s = -V_{\text{out}}\left(n + \frac{1}{2}\right)C_f$$

$$V_{\text{out}}\left(n + \frac{1}{2}\right) = \frac{C_s}{C_f} V_{\text{in}}(n).$$

Thus, the ideal output is simply a scaled and delayed version of the input.

THA Noise Analysis

Now that we have found a way to describe the signal propagation through the THA, we are ready to tackle the detailed noise analysis. To keep this task manageable, we will once again consider the circuit in its two phases and later combine the results. Let's start with the tracking phase.

A circuit model for noise analysis during tracking is shown in Figure 6, which contains only the relevant circuit elements and noise sources that are involved in this phase. The sampling of the thermal noise

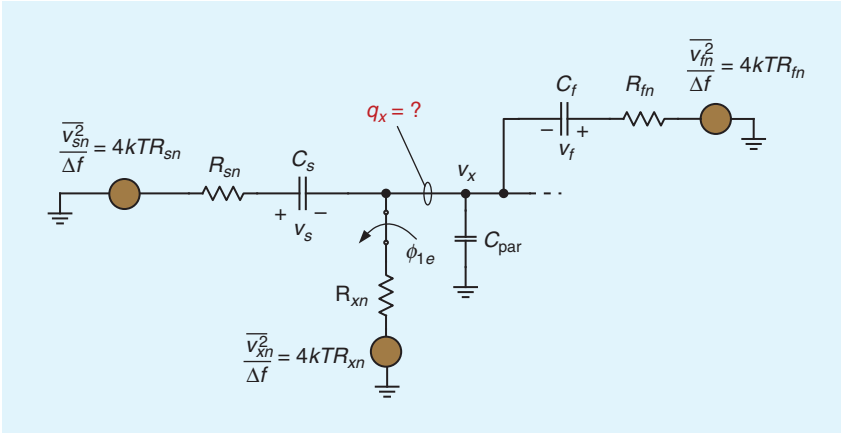


FIGURE 6: Circuit model for noise analysis during ϕ_1 .

occurs when switch ϕ_{1e} turns off. Similar to the way this happens for the signal, the opening of this switch freezes the total charge at node v_x , and this frozen charge is then redistributed onto the feedback capacitor in the following clock phase. So, what we are primarily interested in is finding the total noise charge q_x as indicated. How can this be done?

If we attack this question from the basic principles of electrical engineering, things get quite complicated, but let's give it a try. The total charge at node v_x is

$$q_x = C_s v_s + C_f v_f + C_{\text{par}} v_x.$$

The voltages v_s , v_f , and v_x fluctuate randomly (but in a correlated manner) due to the three noise sources v_{sn} , v_{fn} , and v_{xn} . Specifically, there exist three transfer functions (NTFs) that lead from each independent noise source to the total charge as dissected above. In principle,

the total noise charge could be found by summing the integrals over these NTFs:

$$\begin{aligned} \overline{q_x^2} = & \int_0^\infty 4kTR_{sn} \left| C_s \frac{v_s(s)}{v_{sn}(s)} + C_f \frac{v_f(s)}{v_{sn}(s)} \right. \\ & + C_{\text{par}} \frac{v_x(s)}{v_{sn}(s)} \left. \right|^2 df \\ & + \int_0^\infty 4kTR_{fn} \left| C_s \frac{v_s(s)}{v_{fn}(s)} + C_f \frac{v_f(s)}{v_{fn}(s)} \right. \\ & + C_{\text{par}} \frac{v_x(s)}{v_{fn}(s)} \left. \right|^2 df \\ & + \int_0^\infty 4kTR_{xn} \left| C_s \frac{v_s(s)}{v_{xn}(s)} + C_f \frac{v_f(s)}{v_{xn}(s)} \right. \\ & + C_{\text{par}} \frac{v_x(s)}{v_{xn}(s)} \left. \right|^2 df. \end{aligned}$$

If we went ahead and solved these integrals we would find to our surprise

$$\overline{q_x^2} = kT(C_s + C_f + C_{\text{par}}).$$

Obviously, this elegant and simple result hints that we took a cumbersome

route to get here. It turns out that, once again, the equipartition theorem could have saved us from running integrals. To see this, think about the circuit in the "frozen" state after ϕ_{1e} has gone low. The plates of the three capacitors are connected together, which means that they are connected in parallel from an energy storage perspective. Charges can move from one capacitor to the other, but the sum of charges, and therefore the stored noise energy, is fixed. The equipartition theorem therefore applies to the parallel combination of the three capacitors and using the fact that the energy on a capacitor is $(1/2)Q^2/C$, we can write

$$\frac{1}{2} \frac{\overline{q_x^2}}{C_s + C_f + C_{\text{par}}} = \frac{kT}{2},$$

which directly leads to the above-stated result.

Next, let us think about the hold phase (ϕ_2), using the corresponding model with noise sources shown in Figure 7. There are two main events that we need to track. First, the noise charge q_x that was generated in the tracking phase is redistributed onto the feedback capacitance and leads to a noise contribution in the output of

$$\overline{v_{\text{out},1}^2} = \frac{\overline{q_x^2}}{C_f^2} = \frac{kT(C_s + C_f + C_{\text{par}})}{C_f^2}.$$

Furthermore, additional noise is added to the output sample through the involved switches and also the operational amplifier. While there are many options for the realization of the latter, we will use (for simplicity) the noise model of a single-stage operational transconductance amplifier (OTA). For a single stage OTA, the input-referred noise can be modeled as shown in Figure 7. The excess noise factor α captures details on how the OTA is built, and specifically how much noise is added by the active loads, etc. Typical values for α lie in the range of 1...2.

Now, instead of analyzing the circuit of Figure 7 using brute-force

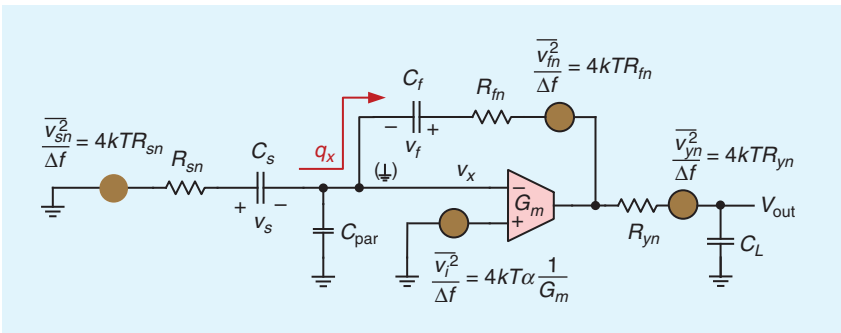


FIGURE 7: Circuit model for noise analysis during ϕ_2 .

algebra, we will start by making an important simplification regarding the frequency response of the network. In particular, we will assume that the resistances R_{sn} , R_{fn} , and R_{yn} do not significantly affect the frequency response of the circuit. This assumption is in line with what a high-speed circuit designer will often shoot for in practice. In fact, it would be quite wasteful to build a circuit whose settling speed is limited by the switches, rather than the speed of the OTA. It usually takes more power to speed up the OTA than to speed up the switches (via W/L sizing of the MOSFETS). With this assumption in mind, the circuit can be simplified as shown in Figure 8.

With this change, the circuit is now a first-order system that can be easily analyzed in terms of its noise transfer functions. Let us begin by finding the NTF from the noise source v_{sn} to the output. Ignoring finite OTA gain (for simplicity), the low frequency NTF is simply $-C_s/C_f$ (the low-frequency closed-loop gain from the inverting input). At high frequencies, the NTF contains a pole that is associated with the equivalent resistance (R_{eq}) and capacitance (C_{eq}) at the OTA output. Both of these terms can be identified as

$$R_{eq} = \frac{1}{\beta G_m}$$

$$C_{eq} = C_L + (1 - \beta)C_f,$$

where $\beta = C_f/(C_f + C_s + C_{par})$ is the circuit's feedback factor. With these ingredients, we can write the contribution of v_{sn} to the sampled output noise as

$$\begin{aligned} \overline{v_{out,sn}^2} &= \int_0^\infty 4kTR_{sn} \left(\frac{C_s}{C_f} \right)^2 \\ &\quad \times \left| \frac{1}{1 + j2\pi f R_{eq} C_{eq}} \right|^2 df \\ &= 4kTR_{sn} \left(\frac{C_s}{C_f} \right)^2 \frac{1}{4R_{eq} C_{eq}} \\ &= \frac{R_{sn}}{R_{eq}} \left(\frac{C_s}{C_f} \right)^2 \frac{kT}{C_{eq}}. \end{aligned}$$

Using the same approach, we can also find the noise contribution

from the OTA. The main difference is that the low frequency noise gain is now set by the noninverting closed-loop expression

$$\begin{aligned} \overline{v_{out,OTA}^2} &= \int_0^\infty 4kT\alpha \frac{1}{G_m} \left(\frac{C_s + C_f + C_{par}}{C_f} \right)^2 \\ &\quad \times \left| \frac{1}{1 + j2\pi f R_{eq} C_{eq}} \right|^2 df \\ &= 4kT\alpha \frac{1}{G_m} \frac{1}{\beta^2} \frac{1}{4R_{eq} C_{eq}} \\ &= \frac{\alpha}{\beta} \frac{kT}{C_{eq}}. \end{aligned}$$

Before moving on to the other noise contributors, it is useful to inspect the ratio of the two terms we have already identified

$$\begin{aligned} \frac{\overline{v_{out,sn}^2}}{\overline{v_{out,OTA}^2}} &= \frac{\frac{R_{sn}}{R_{eq}} \left(\frac{C_s}{C_f} \right)^2 \frac{kT}{C_{eq}}}{\frac{\alpha}{\beta} \frac{kT}{C_{eq}}} \\ &= \frac{R_{sn}}{1/G_m} \frac{\left(\frac{C_s}{C_s + C_f + C_{par}} \right)^2}{\alpha} \\ &\approx \frac{R_{sn}}{1/G_m}. \end{aligned}$$

This ratio says that the noise contribution from the switch in the input branch (R_{sn}) can be minimized by making its on-resistance significantly smaller than $1/G_m$, the reciprocal of the OTA's transconductance. This is precisely what a designer will typically do in practice. As we have already argued, lowering the resistance of a switch is "cheaper" than demanding extra G_m from an OTA. The reader can prove that this argument also applies to the other switch resistances in the circuit. Consequently, we can ignore all but the noise con-

tribution from the OTA in a first-cut noise analysis and write

$$\overline{v_{out,2}^2} \approx \overline{v_{out,OTA}^2} = \frac{\alpha}{\beta} \frac{kT}{C_{eq}}.$$

Whenever we are interested in finding the excess noise due to the switches, we can run a full-blown circuit simulation and compare against the above estimate.

Finally, we are now at a point where we can complete the analysis by combining the noise sources from the two phases

$$\begin{aligned} \overline{v_{out}^2} &= \overline{v_{out,1}^2} + \overline{v_{out,2}^2} \\ &\approx \frac{kT(C_s + C_f + C_{par})}{C_f^2} + \frac{\alpha}{\beta} \frac{kT}{C_{eq}}. \end{aligned}$$

Alternatively, the result can be referred to the input by dividing with the square of the THA's voltage gain

$$\begin{aligned} \overline{v_{in}^2} &= \frac{\overline{v_{out}^2}}{(C_s/C_f)^2} \\ &\approx \frac{kT}{C_s} \left(1 + \frac{C_f + C_{par}}{C_s} \right) \\ &\quad + \left(\frac{C_f}{C_s} \right)^2 \frac{\alpha}{\beta} \frac{kT}{C_{eq}}. \end{aligned}$$

An important observation from this final result is that the input referred noise from the tracking phase is not simply equal to kT/C_s , as often claimed in literature. In some cases, especially when excessive shielding is used at the amplifier input node, C_{par} can lead to a noticeable degradation in noise performance. In addition, note that for a fully differential circuit the predicted noise values must be doubled to account

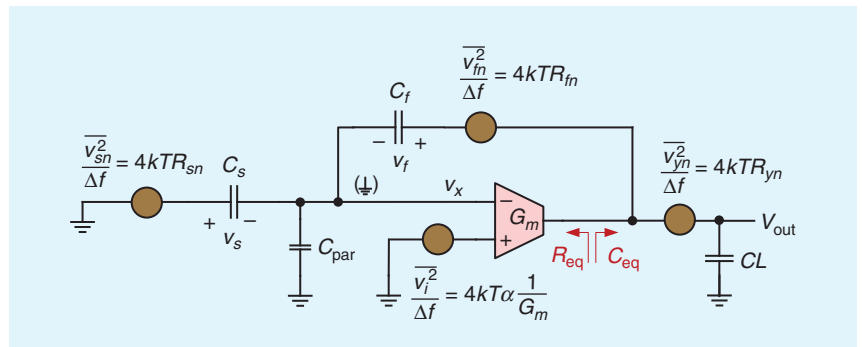


FIGURE 8: Simplified circuit model for noise analysis during ϕ_2 .

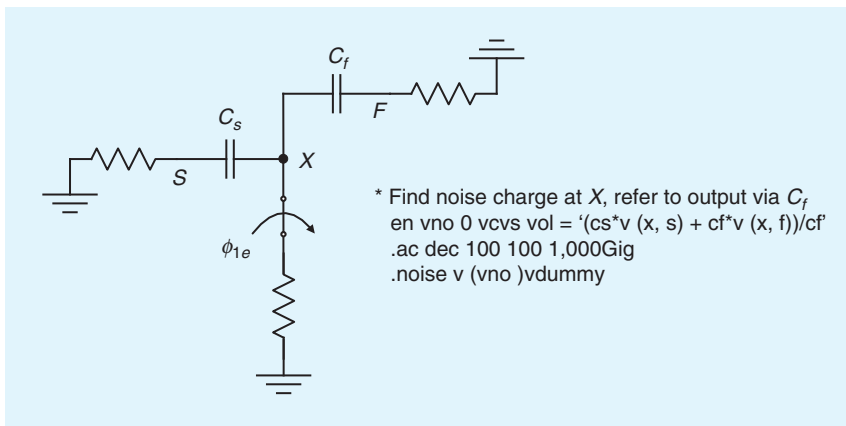


FIGURE 9: Computing the output referred track mode noise using standard .ac/.noise analysis.

for the contribution from both half circuits.

Computer Simulation

Various approaches exist to evaluate the noise performance of SC circuits using simulation tools. In this section we will look at the most commonly used methods: .ac/.noise, pss/pnoise, and transient noise. As shown in [5], all of these methods give accurate results (within a few percentage points) for the discussed THA circuit when properly set up.

The most basic way to simulate thermal noise in SC circuits is to run the readily available .ac/.noise small-signal analyses in a standard SPICE simulator. For the case of the presented THA circuit, we would use the following procedure:

- 1) Turn on the appropriate switches to place the circuit into the hold mode. Run a .ac/.noise analysis, plot the PSD at the circuit output and integrate over all frequencies. This gives the $\overline{v_{out,2}^2}$ component calculated above.

- 2) Turn on the appropriate switches to place the circuit into the track mode. Run a .ac/.noise analysis, compute the noise charge at v_x integrated over all frequencies, and refer to the output. This gives the $\overline{v_{out,1}^2}$ noise component.

- 3) Combine the two noise contributions manually to arrive at the total noise in the output sample.

Since SPICE simulators usually do not output noise charges, we must apply some tricks in performing step 2. This is illustrated in Figure 9 using a simple example. Here, the noise charge is computed using a behavioral source that computes the capacitance-voltage (CV) products of the two capacitors and converts this charge to the corresponding output voltage (after redistribution) by dividing through C_f .

While the outlined .ac/.noise method works perfectly for the discussed THA circuit, it is difficult to apply when the noise sources from the two clock phases are correlated, as it might be the case in a filtering circuit. One way to cover this more general case is via a pss/pnoise analysis.

The acronyms pss and pnoise stand for periodic steady-state and periodic noise [12]. In this simulation approach, the pss analysis is run to capture the steady-state operating point of the circuit when switched periodically between its two phases. The resulting solution is used to compute all the noise NTFs and the desired output noise quantity.

The overall simulation flow can be summarized as follows (for Cadence or a similar EDA environment):

- 1) Set up and run pss analysis by specifying the circuit's clock frequency in the field named "beat frequency" (see Figure 10). In addition, one must pay attention to the parameter "maxacfreq," which specifies up to which frequency the pss result will be valid. Since noise aliases from very high frequencies, it is important

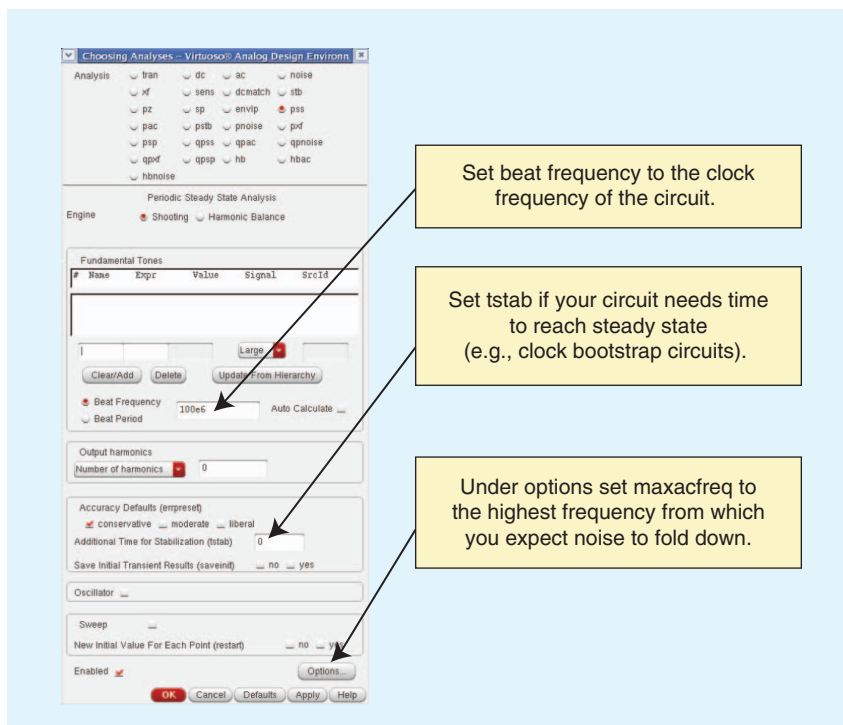


FIGURE 10: Setup for periodic steady-state (pss) analysis.

to set this parameter properly (see [5] and [12]). Finally the parameter “tstab” can be set to achieve convergence for circuits that take time to reach steady-state (useful, e.g., for bootstrapped sampling switches).

- 2) Set up and run pnoise analysis (see Figure 11). To obtain the value of the sampled noise, the analysis variant “timedomain” should be used along with the specification of the desired sample time. As in step 1, we must make sure that the noise is being looked at up to very high frequencies. This is achieved by properly setting the parameter “maximum sidebands” typically to values of 100 or more (see [5] and [12]).
- 3) Plot the sampled noise PSD from $0 \dots f_s/2$ (the result is typically “white”). Integrate the PSD from $0 \dots f_s/2$ to obtain the total sample noise.

As evident from the above description, the pss/pnoise analysis technique requires carefully adjusted parameter settings to perform as desired. Experienced designers will typically run multiple simulations with increasing “max-acfreq” or “maximum sidebands” to validate the proper minimum settings. Fortunately, leading-edge tools have found ways to remove the constraints on sideband count and thereby require simpler setup and deliver significant simulation speed-up [15].

The last option we will consider here is transient noise simulation. This is perhaps the most intuitive method, since it simply relies on injecting transient noise waveforms for each noise source present in the circuit. The transient noise waveforms are automatically generated by the tool and superimposed onto the noiseless transient waveforms. For instance, for a resistor R , the simulator creates a noise voltage waveform with spectral density $4kTR$ and connects this voltage in series. While this method has been available in the ELDO simulator since the

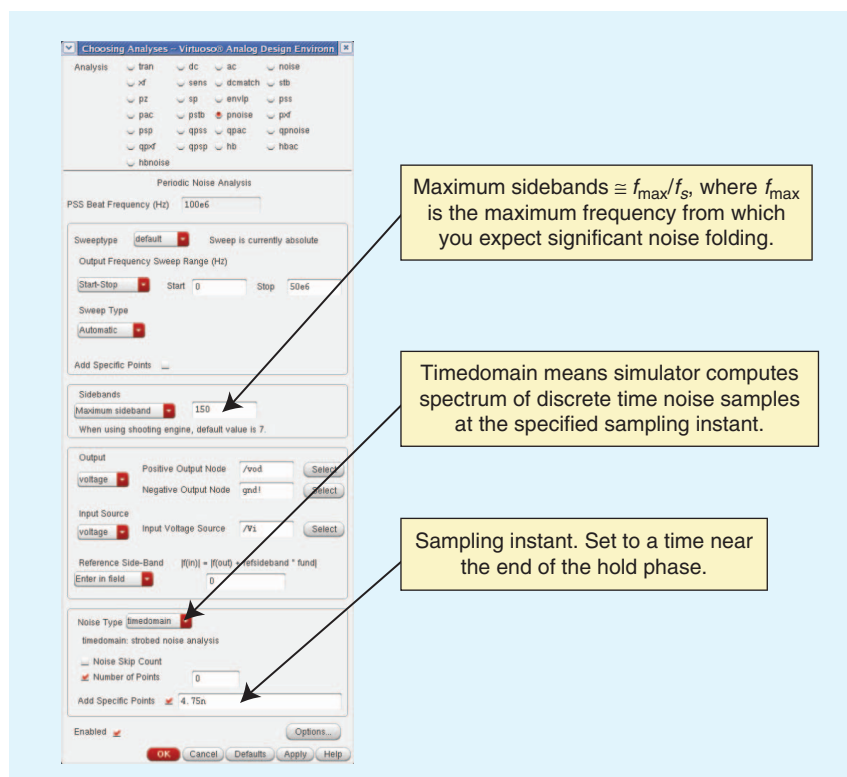


FIGURE 11: Setup for periodic noise (pnoise) analysis.

early 1990s [16], it has become popular for larger circuits only recently. Over the past decade, tremendous progress has been made in the speed and convergence performance of transient simulators in general, and this has also delivered fast transient noise tools that can handle very large circuits at reasonable simulation times [17]. An additional ben-

efit of transient noise simulation is that the circuit timing does not need be periodic, as required in the pss/pnoise approach, for example.

Transient noise simulations are relatively easy to set up. The two most important settings relate to the maximum and minimum noise frequencies of interest. All other settings are similar to the conventional

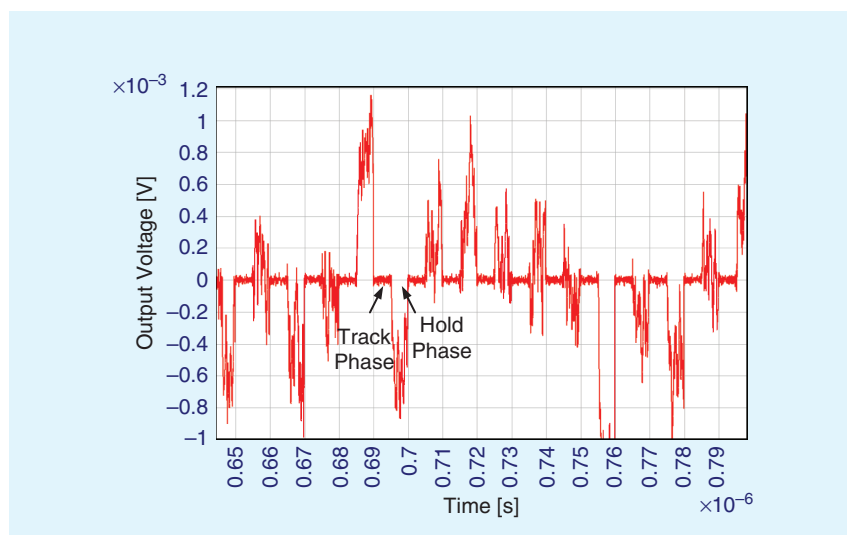


FIGURE 12: Typical transient noise output waveform for a THA.

transient analysis that most designers are familiar with. An example transient noise output voltage waveform for a THA is shown in Figure 12. The quiet periods of the waveform correspond to the track phase when the amplifier output is reset. In the hold mode time slots, a larger variance due to the sampled track mode noise charge and the superimposed OTA noise is apparent. To estimate the variance of the noise samples, one can run a simulation over many cycles (e.g., 1,000), and compute the sample variance of the output near the end of the hold phase.

Summary and Extensions

This article reviewed hand analysis and simulation techniques for the estimation of thermal noise in a charge-redistribution THA. We can summarize some of the key take-homes as follows.

- Analyzing sampled thermal noise usually leads to expressions of the form $m \cdot kT/C$, where the factor m is unity for the elementary track-and-hold.
- The spectrum of sampled thermal noise is usually white, unless incomplete settling or reset creates sample-to-sample correlations.
- Proper noise analysis in charge redistribution circuits requires a charge-domain estimation of the track mode noise; rule-of-thumb voltage noise expressions can lead to inaccurate results.
- The hold-mode noise of most practical THAs is set by the noise of the OTA; the switch noise can be minimized by proper on-resistance scaling.
- Transient noise simulation has become an attractive method to consider due to recent progress in simulator technology. This method also tends to be the most intuitive among all options and applies equally well to nonperiodic circuits.

Several extensions to the presented analysis material can be found in literature. For an analysis of the track mode noise using more

sophisticated two-stage amplifiers, the reader is referred to [18]. Furthermore, Schreier's paper [4] looks more closely at the noise for an integrator configuration and also takes switch noise into account during charge redistribution. In addition, [4] proposes a systematic approach to incorporate the obtained noise expressions into the analysis of a complete functional block, such as a delta-sigma modulator or SC filter.

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